

Module 5 – Assignment Questions

1. What are the key features of Wi-Fi 6, 6E and 7 and how do they differ from previous standards like Wi-Fi 5 (802.11ac)?

1. Wi-Fi 6 (802.11ax): The Efficiency Generation

Wi-Fi 6 was designed primarily to solve the problem of "congested" networks (like stadiums, offices, or homes with dozens of IoT devices).

- **OFDMA (Orthogonal Frequency Division Multiple Access):** Instead of one user taking the whole channel for every transmission, the channel is carved into small "Resource Units" (RUs). This allows the AP to talk to multiple devices simultaneously.
- **Target Wake Time (TWT):** Allows devices to negotiate when and how often they wake up to send or receive data, significantly extending the battery life of smartphones and IoT sensors.
- **1024-QAM:** Increases the "bits per symbol" from 8 to 10, resulting in a 25% raw speed boost over Wi-Fi 5.
- **BSS Coloring:** Identifies signals from neighboring networks with a "color" code, allowing devices to ignore interference from the neighbor's Wi-Fi and stay on their own "lane."

2. Wi-Fi 6E: The 6 GHz Frontier

Wi-Fi 6E has all the features of Wi-Fi 6, but it adds a massive new "highway" for data.

- **The 6 GHz Band:** This is the first time in 20 years that Wi-Fi has been given new spectrum. It adds **1,200 MHz** of new space—triple the amount available in 2.4 and 5 GHz combined.
- **No Legacy Interference:** Older Wi-Fi 4/5 devices cannot access the 6 GHz band. This means 6E is a "pristine" environment with zero congestion from older hardware.
- **Wider Channels:** It supports up to seven non-overlapping **160 MHz** channels, which are necessary for gigabit-speed wireless.

3. Wi-Fi 7 (802.11be): Extremely High Throughput

Wi-Fi 7 is designed for ultra-low latency applications like 8K video streaming and wireless Virtual Reality (VR).

- **320 MHz Channels:** Doubles the maximum channel width of Wi-Fi 6/6E, effectively doubling the speed.
- **MLO (Multi-Link Operation):** A device can connect to the 5 GHz and 6 GHz bands **simultaneously**. In previous standards, a device had to pick one. This drastically reduces latency and increases reliability.
- **4096-QAM (4K-QAM):** Packs 12 bits into every symbol, providing another 20% speed increase over Wi-Fi 6.

- **Multi-RU Puncturing:** If a portion of a wide channel is busy with interference, Wi-Fi 7 can "punch a hole" in the channel and use the rest of it, whereas Wi-Fi 6 would have to abandon the entire wide channel.

2. Explain the role of OFDMA in Wi-Fi 6 and how it improves network efficiency.

- **Role:** In Wi-Fi 6, OFDMA divides a wireless channel into smaller sub-channels called Resource Units (RUs). The Access Point can then allocate these RUs to different clients *simultaneously* in a single transmission opportunity (downlink or uplink).
- **Improvement:** Prior to OFDMA (in previous standards), a single device would often occupy an *entire* 20 MHz or wider channel, even for small data packets (like web requests or VoIP). This was inefficient, like a huge delivery truck carrying only one small envelope. OFDMA solves this by allowing the channel to act like a multi-lane highway or a delivery van with many compartments, transmitting data from/to multiple devices at once. This drastically reduces latency, improves overall throughput in busy networks, and makes better use of available spectrum.

3. Discuss the benefits of Target Wake Time (TWT) in Wi-Fi 6 for IoT devices.

- **Benefits:** TWT allows devices to negotiate specific times and durations for waking up to send or receive data, instead of constantly listening to the beacon for their turn. This enables:
 - **Significantly extended battery life:** IoT devices can remain in a low-power "sleep" state for much longer intervals, crucial for battery-operated sensors and smart home devices.
 - **Reduced network contention:** Since devices know exactly when to wake up, they don't all compete for the channel at once, improving overall network efficiency.
 - **Better performance for critical devices:** Important devices can be scheduled to wake up precisely when needed.

4. Explain the significance of the 6 GHz frequency band in Wi-Fi 6E.

- **Significance:** The addition of the 6 GHz band (up to 1200 MHz of *new* unlicensed spectrum in some regions) is revolutionary because:
 - **It provides "greenfield" spectrum:** The 2.4 GHz and 5 GHz bands are extremely crowded with older Wi-Fi, Bluetooth, microwave ovens, etc. The 6 GHz band is pristine, with virtually zero non-Wi-Fi 6E interference.
 - **It supports wide channels:** The massive amount of contiguous spectrum makes it easy to support multiple 802.11ax/be channels that are 160 MHz (for Wi-Fi 6E) or even 320 MHz (for Wi-Fi 7) wide without overlap, enabling ultra-fast speeds and high capacity.
 - **Mandatory WPA3 security:** Devices using the 6 GHz band must support WPA3, raising the baseline security standard.

5. Compare and contrast Wi-Fi 6 and Wi-Fi 6E in terms of range, bandwidth, and interference.

- **Range:**
 - **Wi-Fi 6:** Operates in 2.4 GHz (better range) and 5 GHz (shorter range). Overall range is generally comparable to Wi-Fi 5.

- **Wi-Fi 6E:** Operates *only* in 6 GHz. The 6 GHz signal will have a slightly *shorter range* and less ability to penetrate walls than 5 GHz, and significantly less than 2.4 GHz.
- **Bandwidth:**
- **Wi-Fi 6:** Shares existing 2.4 GHz and 5 GHz bands, often heavily congested. Supports channel widths up to 160 MHz, but practical 160 MHz usage can be limited by DFS requirements in 5 GHz.
- **Wi-Fi 6E:** Accesses *new, massive* spectrum (up to 1200 MHz), dedicated purely to 6E. This allows for numerous non-overlapping 802.11ax/be channels, including multiple 160 MHz (6E) or even 320 MHz (Wi-Fi 7) channels, significantly increasing available bandwidth and capacity.
- **Interference:**
- **Wi-Fi 6:** Still faces significant interference from legacy Wi-Fi (a, b, g, n, ac), microwaves, other 2.4/5 GHz devices. Technologies like BSS coloring help, but the underlying bands are busy.
- **Wi-Fi 6E:** Has effectively *zero interference* from non-Wi-Fi 6E devices or older Wi-Fi versions (which cannot operate in 6 GHz). The band is inherently quiet, leading to much cleaner signals and vastly improved performance reliability.

6. What are the major innovations introduced in Wi-Fi 7 (802.11be)?

- **Extremely High Throughput (EHT):** Aims for peak theoretical speeds up to 46 Gbps.
- **320 MHz Channels:** Doubles the maximum channel width compared to Wi-Fi 6E (160 MHz), further boosting speed.
- **Multi-Link Operation (MLO):** Allows a single device to connect to an AP using multiple frequency bands (e.g., 2.4, 5, and 6 GHz) and switch or combine them for higher speed, reliability, and reduced latency.
- **4096-QAM (4K QAM):** Encodes 12 bits of data per symbol (vs 10 bits in 1024-QAM), providing a 20% data rate boost in optimal signal conditions.
- **Multi-RU (Resource Unit) support:** Improves upon OFDMA by allowing a single device to be allocated multiple different-sized RUs for even more efficient bandwidth usage.
- **Reduced Latency:** MLO and other EHT features are specifically designed to minimize delay, critical for applications like real-time gaming, VR/AR, and video conferencing.

7. Explain the concept of Multi-Link Operation (MLO) and its impact on throughput and latency.

- **Concept:** MLO is a groundbreaking feature where a single MLO-capable device and AP can establish multiple independent logical wireless links across different frequency bands (e.g., one link on 5 GHz and another on 6 GHz). They can then use these links in various ways:
 - **Load Balancing / Link Steering:** Move traffic seamlessly to the clearest link, reducing congestion.
 - **Aggregation:** Combine the links for higher simultaneous throughput (e.g., splitting a file across two bands).
 - **Reliability:** Send identical data over multiple links to guarantee delivery and reduce packet loss.
- **Impact on Throughput:** MLO can significantly *increase* throughput by allowing devices to utilize the capacity of multiple bands concurrently, rather than being limited to just one at a time.

- **Impact on Latency:** MLO *dramatically reduces latency and jitter*. Devices don't have to wait for contention on one single band to clear. If one link is busy, the critical packet can be immediately transmitted over an alternative, idle link, ensuring deterministic, low-latency performance essential for time-sensitive applications.

8. What is the purpose of 802.11k and v, and how does it aid in roaming?

- **Purpose of 802.11k (Neighbor Reports):** Helps mobile devices (clients) scan for better neighboring Access Points (APs) *before* they need to roam. The currently connected AP can provide the client with a "neighbor list" of other available APs, along with their channel and link quality information.
- **Purpose of 802.11v (Wireless Network Management):** Allows the overall Wi-Fi network to proactively advise connected clients on the *best* AP to associate with, rather than letting the client independently decide. APs can even "force" a client to roam to another AP for load balancing or when its link quality degrades significantly.
- **Aid in Roaming:** 802.11k allows clients to build a prioritized list of candidate APs, speeding up the scanning phase of roaming. 802.11v provides network-directed intelligence, ensuring clients connect to APs that offer the best performance and that the overall network remains balanced and efficient. This collaborative approach leads to much faster, more intelligent, and less disruptive roaming decisions.

9. Explain the concept of Fast BSS Transition (802.11r) and its benefit in mobile environments.

- **Concept:** 802.11r is a standard designed to significantly *reduce* the time it takes for a pre-authenticated mobile device to re-authenticate and re-associate with a new Access Point during a roam. In standard WPA2/WPA3 enterprise environments, full re-authentication is a complex, time-consuming process involving the EAP server. 802.11r achieves "Fast Transition" (FT) by using a distributed key caching mechanism. Before roaming, the client can pre-negotiate authentication material with the target AP (or have it transferred from the old AP), eliminating most of the back-and-forth round trips to the authentication server at the moment of the roam itself.
- **Benefit:** The crucial benefit is maintaining connectivity and quality for *latency-sensitive applications* while moving. Without 802.11r, a roam might cause a dropped voice call or a freezing video call. With 802.11r, the "handoff" is so fast (tens of milliseconds vs. seconds) that users experience virtually no interruption, which is vital for smooth operation in high-density mobile enterprise environments.

10. How do 802.11k/v/r work together to provide seamless roaming in enterprise networks?

- **The Seamless Roaming Workflow:** These three standards work together in a powerful "triple-play" combo:
 - **802.11k (Preparation):** The connected AP gives the client a neighbor list. The client uses this intelligence to scan *efficiently*, identifying suitable AP candidates in advance.
 - **802.11v (Guidance):** The network can monitor overall load and signal conditions. It can then proactively guide the client to the absolute best neighbor AP, ensuring the roam isn't just fast, but *smart* (e.g., for load balancing).
 - **802.11r (Execution):** Once the client (aided by 802.11k) or the network (aided by 802.11v) determines it's time to roam, 802.11r takes over. Because key negotiation is largely done

beforehand, the re-association and full data path setup with the new AP happens extremely quickly, minimizing the data loss/latency spike to a negligible level.

- **Overall Impact:** 802.11k minimizes scanning time, 802.11v ensures optimal AP choice, and 802.11r minimizes the re-authentication gap. The end result is that a user on a critical application (like a VoIP call) can walk across a large campus, switching between dozens of APs, without a single hiccup, ensuring truly seamless mobility in enterprise environments.